Class-Aware Multifaceted Approach for Imbalanced Dermoscopic Image Classification

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***Abstract*—A major health issue globally and a challenging problem to cure, diagnosing melanoma in early stages and accurately is very vital in lowering the deaths. Unlike the deep learning, which has proved to be an effective tool in the automation of the dermoscopic analysis and the analysis of the images of the skin, the deep learning is not always a reliable tool in the clinical setting due to the imbalance in data used to train the models. Such an imbalance in training data that results in overtraining the model to overlook critical features in smaller and minor lesions. In order to deal with the limitation, we present a Class-Aware Multifaceted Approach integrating class-specific data augmentation, a weighted focal loss function and a ResNet50 model structure (ResNet50 + CBAM) with attention mechanisms added to it. This new model on popular skin cancer data which is HAM10000 dataset which outperformed the standard model which is EfficientNet-B0 with a Macro F1-Score of 0.7164 appeared compared to 0.5242 and it can be further proved by the addition of the qualitative analysis with the help of Grad-CAM which shows that the integrated mechanism allowed the model to focus on the critical aspects of the skin lesions and minimized the influence of the background artifacts and thus made it more reliable. This paper has shown that with a holistic integration of fata augmentation, better algorithms, and architectural innovation of smarter designed models to improve the diagnostic accuracy and explainability of automated skin cancer screening systems, one can make better and reliable clinical decision support.**

***Index Terms*—The classification of skin cancer, class imbalance, deep learning, attention mechanism, focal loss, interpretability**

I. Introduction

One of the health problems of significant concern to the society is skin cancer. Melanoma is the most prevalent and hazardous type of skin cancer that results in the greatest death rate and its occurrence is increasing in every corner of the world [1], [2]. Despite the fact that the melanoma can be treated with high survival rate in case it is diagnosed at the early stages, the provided disease can be rather challenging to notice in the first stages of its development. The survival is more than 99 percent in case when the cancer is localized and it was discovered within 5 year however it is just 15-20 percent in case where it is metastatic [1], [3].

Through artificial intelligence (AI) and deep learning, especially convolutional neural networks (CNNs) to improve the diagnostic abilities. These are the devices used by individuals to help dermatologists to evaluate the dermoscopic photographs [4], [5]. TThe instruments are partial facilitators of the workload and more precise in the diagnosis especially in the situations that have limited specialists. The massive problem that prevents the complete utility of AI is that in a significant majority of data sets, as with HAM10000 [6], the proportions between the categories will not be equal, and the number of cases of benign lesions will be much larger than the number of cases of melanoma [7]. This skew causes the common class to be favourable to the model. Better put, it ignores the situations of harmful minorities and that can never be applied in clinical practice [7], [8].

This paper will aim at defeating this obstacle by proposing a Class-Aware Multifaceted Approach. Not only will it improve the overall accuracy but also regulate data and learning strategies in a quantity of levels. Within the data, algorithm, and architecture, we are concerned with the problem of class so that we ascertain that the model can detect melanoma with the necessary degree of accuracy and reliability.

A. Related Work

The fact that the data we employ is not always balanced is one of the greatest obstacles in the classification of skin diseases. To address this, specialists tend to either clean up data, modify the math or the system architecture presented in Table I. As an example, a paper by Alsaidi et al. [7] compared the existing data boosting algorithms to GANs in order to determine which one was more suitable in bridging the gaps in the smaller yet less prevalent sets of diseases. Other studies including the works by Cui et al. [9] and Park et al. [10], have resorted to hybrid solutions, or reconfigured multi-task learning models to enhance feature learning. Other more recent methods include Meta-learning by Kumar et al. [11] and Adaptive sampling by Nguyen et al. [8] which strive to make learning more intelligent. Both approaches continue to possess some constraints including the complexity of model training, excessive consumption of computational resources or the absence of a process that can explicitly clarify the findings. This study will be used to address these gaps.

B. Gap Analysis

As we have seen, even though there are so many developed techniques in the market, there are great gaps in the field. Traditionally, people resort to generic appraisal methods that do not actually portray the diversity of the issue especially when minority classes are concerned and their visual specifics are interacting [7]. TThe gap in the historical research is

TABLE I: Literature review of imbalanced skin lesion classification techniques

Author	Dataset	Level	Technique	Results	Limitations
Alsaidi et al. (2023) [7]	HAM10000	Data	Basic augmentation, GANs	Basic augmentation better than GANs	Lacks class-specific handling strategy
Dildar et al. (2021) [12]	ISIC 2018 / 2019	Data	Geometric Augmentation	Improved accuracy via balanced distribution	Uses generic augmentation, potentially distorting lesion features
Srinivasu et al. (2021) [13]	HAM10000	Architecture	MobileNet V2 + LSTM	Low computational cost with good accuracy	Lightweight model may miss fine-grained texture details
Punn & Agarwal (2021) [14]	HAM10000	Architecture	InceptionV3 + Attention	Attention improved feature focus significantly	High model complexity and training time
Cui et al. (2025) [9]	ISIC-style datasets	Hybrid	Augmentation + Attention + Loss	Multi-technique integration yields good results	Focused on segmentation and was not tested on HAM10000
Kumar et al. (2024) [11]	SD-198 / ISIC 2018	Algorithm	Meta-learning, few-shot learning	Learns quickly from small amounts of data	Model is complex and difficult to train
Park et al. (2023) [10]	ISIC 2017 / 2018	Architecture	Multi-task learning	Performance is improved by multitasking	Uses high computing resources
Nguyen et al. (2024) [8]	ISIC 2019 / 2020	Data/Algo	Adaptive sampling	Helps with minority class convergence	Lacks an Attention mechanism to explain the results

that most focused on a hybrid methodology of the image segmentation and hardly any research on standard datasets classification [9]. In addition, most studies evaluate models based on overall accuracy without respect to clinically relevant outcomes e.g. melanoma recall, which is quite a problem with skewed data [7], [8].

C. Problem Statement

Following are the main questions addressed in this study.

1) Is a multifaceted solution capable of increasing the capacity of the system to detect minorities classes but vital conditions, namely melanoma?
2) What advantages does our proposed three stage model have over basic baseline model (EfficientNet-B0)?
3) Why does ResNet50+CBAM model give more visual evidence and qualitative interpretation (Via Grad-CAM) as compared to the standard baseline model in justifying more medical decisions?

D. Novelty of our work

The main originality of the study does not lie in the creation of a new deep-learning algorithm but a new framework that addresses the long-standing problem of class imbalance when analyzing skin lesions data with dermoscopic images. This was done through the coordination of three strategies. To begin with, we apply a policy of augmentation of classes that introduces heterogeneity to samples of the minority lesions without compromising the statistics of the common majority classes. Second, we have enhanced the default ResNet50 backbone with the Convolutional Block Attention Module (CBAM) by adding at each stage. It will enable the model to reject background noise to give attention to the significant background of lesions and features. There is the third, in which we perform focal loss to bias the system to learn hard-to-classify cases of melanoma though the overall performance remains strong, which standard models ignore.

E. Our Solutions

In order to address these issues, we suggest a three-layered design to solve the problem of class- imbalance in the dataset HAM10000. We conduct class specific augmentation to balance the sample distribution by enriching minority classes. Our Focal Loss Function is applied algorithmically and assists the model in learning hard-to-classify, hence. During architecture stage, we append a CBAM attention block to the ResNet 50 backbone in order that the model focuses on meaningful mutilations. We find that our experiments have huge macro F1-score and melanoma recall that is larger than the base model. Grad-Cam is also utilized to give a visual illustration of our decisions, so that the model is guided by the true detail of the lesion and hence enhancing accountability and credibility in clinical practice.

II. Methodology

The current paper adopts experimental research, which was conducted in order to design and train a deep-learning based model combining an attention model and a multi-dimensional data-processing method, to classify skin spots in dermoscopy images.

A. Dataset

The dataset that we used is Human Against Machine with 10,000 training images (HAM10000) [6]. It has 10,015 dermoscopic images of seven different lesion types that include Melanocytic nevi (Nv), Melanoma (Mel), Benign keratosis-like lesions (bkl), Basal cell carcinoma (bcc), Actinic keratoses (akiec), Vascular lesions (vasc), and Dermatofibroma (df) as shown in Figure 2. The data has a serious class imbalance, where a majority (67%) of the cases are Nv, and melanoma only (11%) as shown in Figure 1 .

B. Overall Workflow

There are three workflow stage (see Figure 3). We carry out initially a massive data exploration and preprocessing, such as

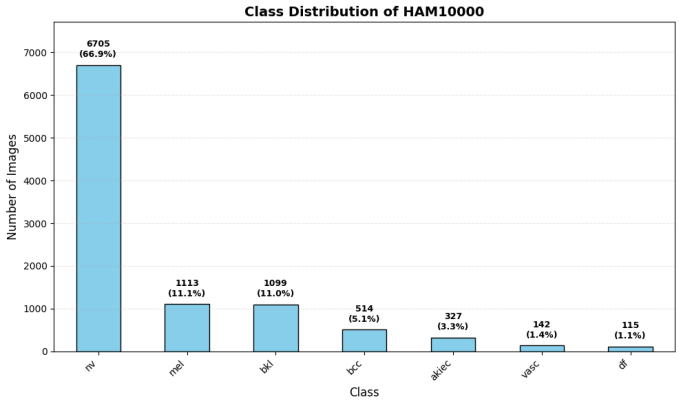


Fig. 1: The class distribution of the HAM10000 dataset.

stratified sampling of HAM10000 and subdividing it into a training set of 70 percent, a validation set of 15 percent, and a test set of 15 percent. This means that all dermatological classes were represented proportionately. Second, we deal with the issue of the imbalance of the classes and create powerful predictive models in three parallel layers.

This paper proposes methods to adjust, at data-level, by augmenting class-specific data via minority lesions resampling (aggressive resampling) and majority subclass resampling (minor resampling; (B) Architectural improvement by placing CBAM on the ResNet-50 backbone to improve feature extraction. (C) Optimization based on an algorithmic approach with focal loss and a two-stage training schedule, where sample misclassification of minority groups is more intensified at the first early fine-tuning stage.

The three layers synergies in order to minimize bias at the expense of radical selectivity. The last process will be an in-depth analysis and practical findings. In correcting the problem of overfitting, we estimate the model efficacy by measuring the effectiveness with accuracy, sensitivity, specificity, and F1- score on the test set. Grad-CAM heatmaps are also used to demonstrate how the model reasoned, which generates transparency and applicability to clinical practice.

C. Data-Level Strategy

Augmentation based on classes was used to address the imbalance of classes in the HAM10000, and generic random sampling was abandoned. In most of the classes (Melanocytic nevi), rotating and flipping horizontally appear to be the light transformation forms we judged and changed so that the natural appearances are retained. In the case of minority classes (melanoma), we used heavy augmentation the wide-range rotations, and brightness changes, and color noise to artificially increase the diversity of the training set and minimize the risk of overfitting [7] as shown in Table III.

D. Architectural-Level Strategy

The suggested architecture is based on ResNet50, which is used as the basic network. We added the convolutional block attention module (CBAM) [15] to improve the ability to distinguish lesions from a noisy background. CBAM operates through two sequential mechanisms. It begins with the channel attention module to identify which features are important by calculating the weight of interest using the equation as formulated in Equation (1).

$$M_c(F) = \sigma\left(\text{MLP}\left(\text{AvgPool}(F)\right) + \text{MLP}\left(\text{MaxPool}(F)\right)\right) \tag{1}$$

where $F \in \mathbb{R}^{C\times H\times W}$ is the input feature map, AvgPool and MaxPool denote global average pooling and max pooling respectively along the spatial dimensions, producing descriptors $\mathbb{R}^{C\times 1\times 1}$. MLP denotes a shared multi-layer perceptron with one hidden layer, and σ is the sigmoid activation function. The resulting attention vector $M_c(F) \in \mathbb{R}^{C\times 1\times 1}$ is applied to F via element-wise multiplication to emphasize informative channels.

After the filter features in the first step. The spatial attention mechanism as shown in Equation (2) identifies important positions in the image. It creates a spatial attention map to identify the position of the lesion and irrelevant background. This process helps to make the model more robust against noise in dermoscopy images.

$$M_s(F') = \sigma\left(f^{7\times 7}\left([\text{AvgPool}(F'); \text{MaxPool}(F')]\right)\right) \tag{2}$$

where $F' \in \mathbb{R}^{C\times H\times W}$ is the channel-refined feature map, $[\cdot;\cdot]$ denotes channel-wise concatenation, $f^{7\times 7}$ represents a convolutional layer with a 7×7 kernel, and σ is the sigmoid function. The operations $\text{AvgPool}(F')$ and $\text{MaxPool}(F')$ are performed along the channel dimension, yielding two 2D maps $\mathbb{R}^{1\times H\times W}$. The output $M_s(F') \in \mathbb{R}^{1\times H\times W}$ is a spatial attention map that highlights lesion regions and suppresses background noise.

In the proposed approach, we utilize CBAM at the end of every residual block from layer 1 to layer 4 rather than placing it in single spot. Our design it should be adaptive feature refinement throughout the network from low level textures to high level semantic patterns.

In this paper, we address class imbalance at the algorithmic level by replacing the standard cross-entropy loss with the focal loss function [16]. This strategic substitution reduces the importance of easy example classification and forces the model to focus on hard examples, which is crucial for handling imbalanced dermatological datasets [7]. The function is defined in Eq. (3).

$$\text{FL}(p_t) = -\alpha_t(1-p_t)^{\gamma}\log(p_t) \tag{3}$$

where p_t is the model's estimated probability for the true class, α_t is a weighting factor for class balancing, and γ is the focusing parameter that reduces the relative loss for well-classified examples.

We set the focusing parameter γ according to the class proportion. This value is critical for reducing the gradient contribution from the majority class, Melanocytic Nevi (NV) [6]. Consequently, this forces the model to extract melanoma lesion



Fig. 2: Sample Images per Class with Size Original.

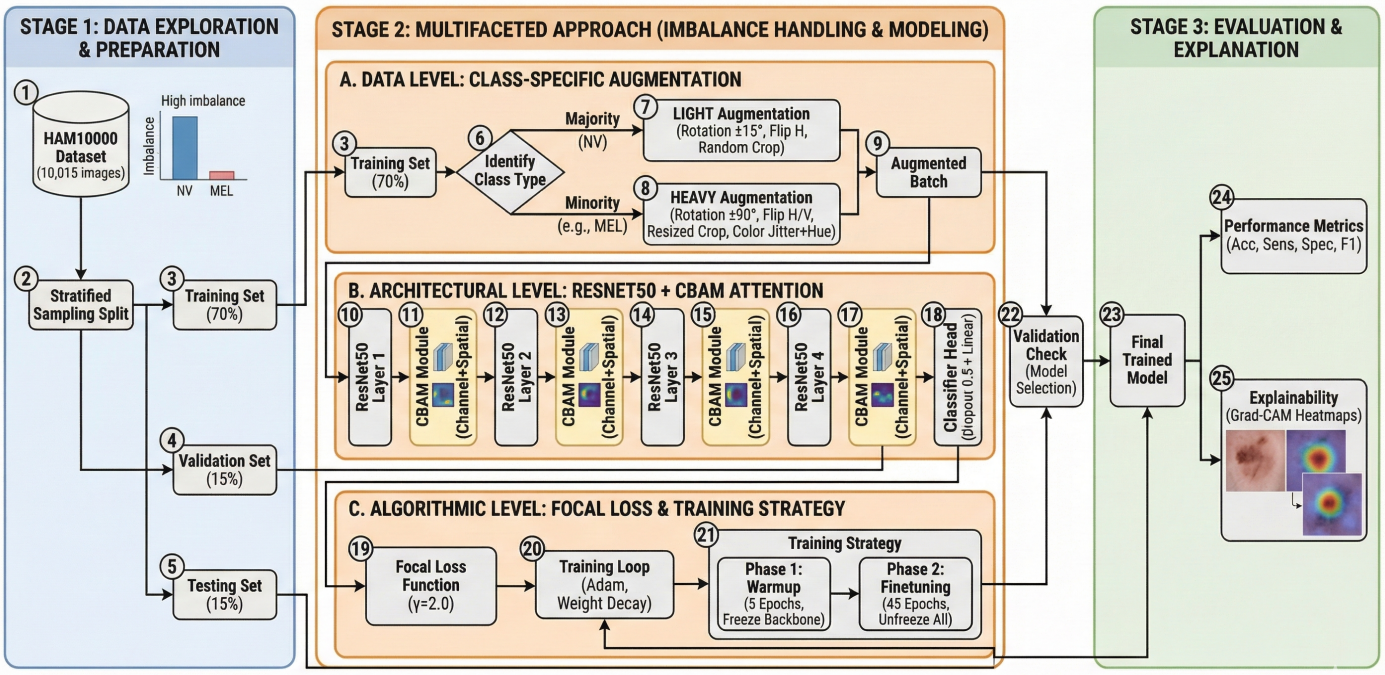


Fig. 3: The overall workflow diagram illustrating the pipeline.

features more effectively without being overwhelmed by the majority class. Complementing the loss function, we employed a two-phase training strategy. The first phase involves a warm-up period of 5 epochs to stabilize the network and prevent pre-trained ImageNet weights [17] from being corrupted. Subsequently, the model enters the fine-tuning phase for 45 epochs by unfreezing all layers, allowing the entire network to fully adapt and learn the sensitive features of skin lesions [9].

E. Experimental Settings

We perform in Python 3.8 environment, PyTorch 1.12 framework, and Albumentations libraries for data augmentation. We divided configuration part in three main as shown in Table below.

1) Hyperparameters configuration: to configure parameters for training the model in two periods 1) warmup and 2) finetuning for the model to adapt well as presented in Table II.
2) Data augmentation strategy: to define different for increasing data, where minority classes are processed using heavy intensive techniques as presented in Table III.
3) Model Architecture configuration: to addjust ResNet50 with adding every CBAM in each of layer. This approach for refining features before sending to next channel attentiion and help the model select important features. Spatial attention help to improve the efficiency of focusing on small as presented in Table IV.

TABLE II: Hyperparameters setting

Hyperparameter	**Value**
Input Image Size	224×224
Batch Size	32
Learning Rate	1×10^{-4}
Weight Decay	3×10^{-5}
Epochs	50 (Warm-up: 5 epochs with frozen backbone, Fine-tune: 45 epochs with all layers unfrozen)
Loss Function	Focal Loss ($\gamma = 2.0$)
Optimizer	Adam

TABLE III: Data augmentation setting

Technique	**Baseline**	**Majority (Light)**	**Minority (Heavy)**
Target Class	EfficientNet-B0	NV	Melanoma
Rotation	$\pm 30°$	$\pm 15°$	$\pm 90°$
Flip	H & V ($p = 0.5$)	H ($p = 0.5$)	H & V ($p = 0.5$)
Crop & Resize	Rand. Crop (224^2)	Rand. Crop (224^2)	Rand. Resized (0.8-1.0)
Color Jitter	Brightness/ Contrast/ Saturation: 0.1	-	Brightness/ Contrast/ Saturation: 0.1, H: 0.01

F. Evaluation Metrics

1) **Quantitative Metrics** To evaluate model performance, we not only consider overall accuracy but also use sensitivity (recall). This measures the accuracy rate of melanoma lesion detection to protect against false negatives. We also assess the reliability of positive prediction (precision). However, the main metric to select the best model is the macro F1-score. It is the harmonic mean that gives equal importance to all classes and addresses the imbalanced data problem.

TABLE IV: Resnet50 model configuration

Parameter	**Value**
Backbone Architecture	ResNet50 (Pretrained on ImageNet)
Attention Module	CBAM (Channel + Spatial)
CBAM Reduction Ratio	16
CBAM Spatial Kernel	7×7
Integration Points	After Layer 1–4
Classifier Head	Dropout (0.5) + Linear Layer

2) **Qualitative Explainability** We applied Gradient-weighted Class Activation Mapping (Grad-CAM) to create a heatmap, which shows the areas the model focuses on during decision-making. This allows us to verify the behavioral correctness of the model by determining whether accurate predictions are based on the actual lesion or artifacts, such as black borders or hairs. This visualization serves as evidence to ensure that the attention mechanism works as theorized.

III. Results

A. Analysis of Learning Curves

We have observed the graph of Figure 5 . The variations of the learning curves of EfficientNet-B0 and Resnet50+CBAM are few, but significant. The most striking aspect of the figure is that there was a significant difference between the loss curve of EfficientNet-B0, i.e. feature learning was not stable. In contrast, ResNet50+CBAM has smooth-varying loss curve which suggests more steady convergence. Also, we found out that the line in the accuracy curve, which depicts the CBAM-enhanced model exhibits gradual improvement in the rise of the curve which has a lesser train-validation gap. In this line, there was increased generalization. It implied that inclusion of the CBAM module is not fruitless in the increasing attention of salient areas and reduction of noise.

B. Comparative Performance Analysis

To measure the relative performance of the architectures under analysis, we tested a group of 1,497 images with three different models the baseline EfficientNetB0, the control ResNet50vanilla, and our adaptation of ResNet50 with the Convolutional Block Attention Module (CBAM). The quantitative outcomes are given in detail in Table V . The best macro F1 -score of 0.7164 with a balanced accuracy of 0.6976 was obtained using the ResNet50 -CBAM configuration. When comparing the ResNet50-CBAM with the baseline Efficient-Net -B0, the latter achieved a macro F1 -score of 0.5242 and a balanced accuracy of 0.6845, which is statistically significant. Conversely, the ResNet50-vanilla that does not have an attention mechanism had the lowest balanced accuracy of 0.6129 and a macro F1-score of 0.5505. In this manner,

the ResNet50-CBAM architecture that was proposed showed a significantly higher performance in all the key metrics.

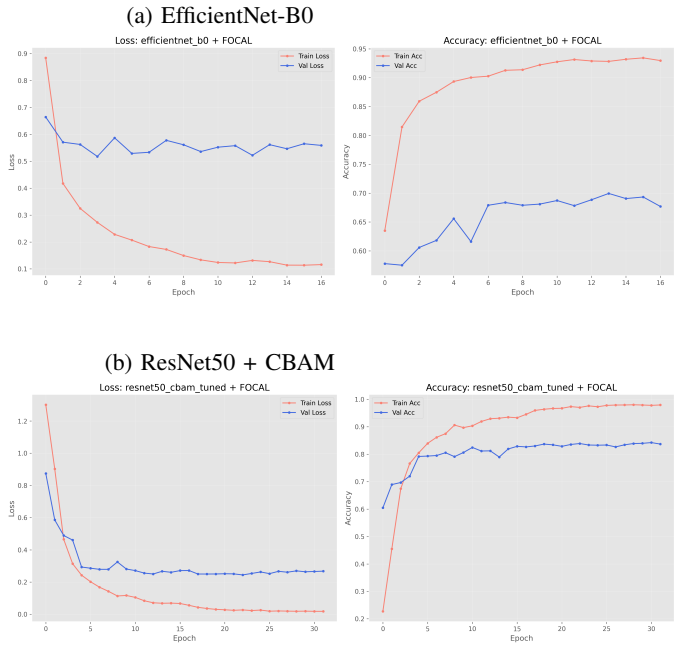


Fig. 4: : Comparison of Grad-CAM visualizations for a correctly classified melanoma case.

TABLE V: Comparative performance metrics of the baseline and proposed models

Model Name	Macro F1	Bal. Acc.	Mel. Prec.	Mel. Rec.	Mel. F1
EffNet-B0 (Baseline)	0.5242	0.6845	0.2687	0.6190	0.3748
ResNet50_ vanilla (Ablation)	0.5505	0.6129	0.5119	0.2560	0.3413
ResNet50+ CBAM (Proposed)	0.7164	0.6976	0.6575	0.5714	0.6115

C. Minority Class Classification Results

Figure 5 shows the classification outcomes of minority classes with special attention on melanoma. That is why the EfficientNet -B0 has the highest melanoma recall at 0.6190 i.e. it hit 104 out of 168 images. However, it is a little messy with accuracy: at least -0.2687, which drops the F1 -score to a mere 0.3748. On the other hand, our ResNet50+CBAM performed very well, recall 0.5714 (96 correct predictions), a little lower on recall and much higher on precision, with an attractive F1-score of 0.6115. Another breakdown of misclassification reveals that ResNet50+CBAM only flipped 55 melanomas to nevus, by far less than 99 flips of ResNet50 vanilla. In other classes of minorities, ResNet50+CBAM achieved an F1 of 0.7417 on basal cell carcinoma (BCC) and 0.6610 on actinic keratoses (AKIEC)

D. Qualitative Analysis via Grad-CAM

Conducted a qualitative analysis with Grad-CAM in order to take a peep of the areas that our models actually relied on to make decisions. Figure 6 that contrasts the baseline and proposed models on a case of successfully classified (true positive) melanoma, the EfficientNet -B0 model was inclined to shoot out fuzzy or spread-out activations (colored in red). These dots used to fall on the noise of background or irrelevant spots. At the same time, ResNet50+CBAM narrowed down to the lesion more precisely and intensely, and this demonstrates that the attention mechanism does actually assist in filtering out superfluous information.

An example of misclassification can be seen in Figure 7 of ResNet50+CBAM, to which an AKIEC (early-stage cancer) was misclassified as a BKL (benign tumor). The Grad-CAM map indicates that the model fired on the right location but the high level of similarity between AKIEC and BKL distracted the focus of the model. This confusion made it inappropriate to extract the finer details to identify the type of disease in the picture correctly.

IV. Discussion

A. The Efficacy of the multifaceted approach

The findings obtained during the present experiment demonstrate that the Class Aware Multifaceted Approach would have acquired a significant advantage of addressing the problem of the overall class imbalance that currently inhabits the field of medical imaging. By surpassing the limitations of traditional uni-modal models through the joint application of data, algorithmic, and architectural interferences, we have shown that we perform better than the base models. These baselines were prone to over-predicate malignancy in order to achieve the high recall rates hence providing artificial results of high false-positive rates rendering them inappropriate in real clinical practice. In order to beat this, our training approach assists in combating this issue by enforcing a more adverse leaning in terms of focal loss and focused augmentation, resulting in a model that not only happens to be educated with the frequencies of classes, but it additionally learns in a discriminative manner.

B. Minority Classes Detection Improvement

The identification of minorities has improved significantly, and it is demonstrated by the fact that the melanoma F1 score was increased almost twice (0.37 to 0.61). A comparison of the proposed model with a vanilla ResNet50 supported by an ablation study indicates the inadequacy of the latter to address the problem of class-imbalance since it made incorrect classifications of 99 benign nevi images. This misclassification error reduced to 55 with accounting of the CBAM attention mechanism depicting the fact that attention modules augment discrimination of small features between benign nevi and early melanoma by the network. However, there are still some residual misclassifications, which implies a weakness of the existing framework. Future research that needs to be conducted must take into account the use of extra data modalities,

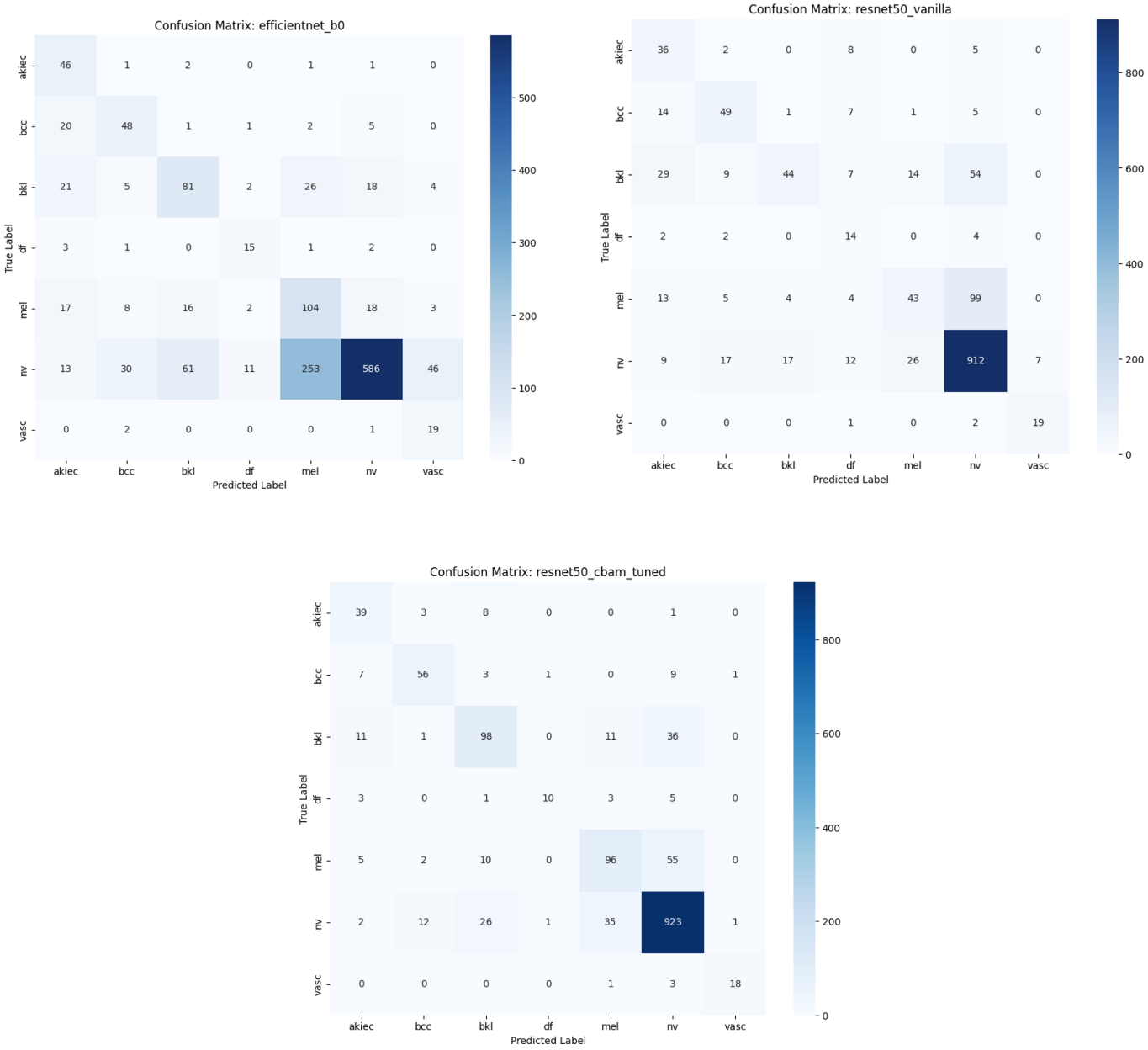


Fig. 5: Comparative confusion matrices of the EfficientNet-B0 (baseline), ResNet50_vanilla, and the proposed ResNet50+CBAM models evaluated on the test set.

including microscopic pictures or detailed clinical records as a means to further narrow the gap between diagnosis and truth.

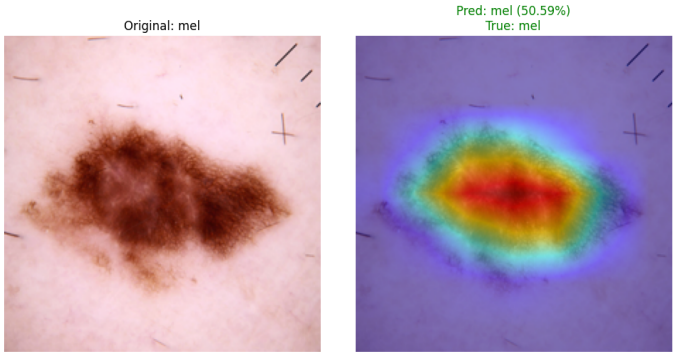
C. Clinical trust and interpretability

Grad-CAM visualisations show that the ResNet50+CBAM model has reasoning behaviour consistent with clinical diagnostic heuristics, in that it integrates within the areas of lesions instead of other distracting artefacts like gel or hair. It is an explanation ability that removes the black-box perception largely part of AI systems. To enable it on a practical use, the validation of the model and its stability in different settings will require impacting on an independent set of people. In addition, investigating active weighting schemes of focal loss parameters may possibly lead to additional declines in residual error rates resulting in clinical confidence in the system.

D. Future Directions

The proposed areas of future research would be to ensure that the performance of the model can be tested based on external independent data to ensure that the model is stable in other environments. Another image of data we are considering adding is the history of patient clinical history to help

(a) EfficientNet-B0



(b) ResNet50 + CBAM

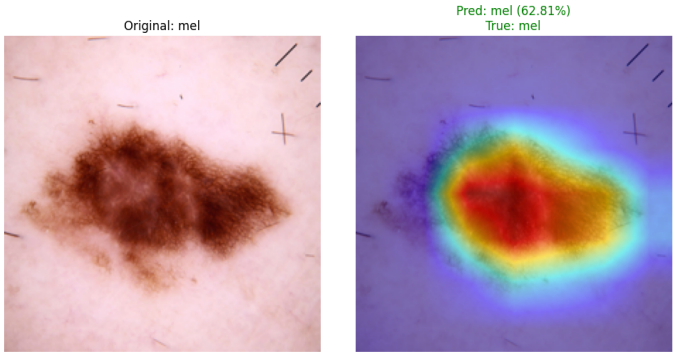


Fig. 6: : Comparison of Grad-CAM visualizations for a correctly classified melanoma case.

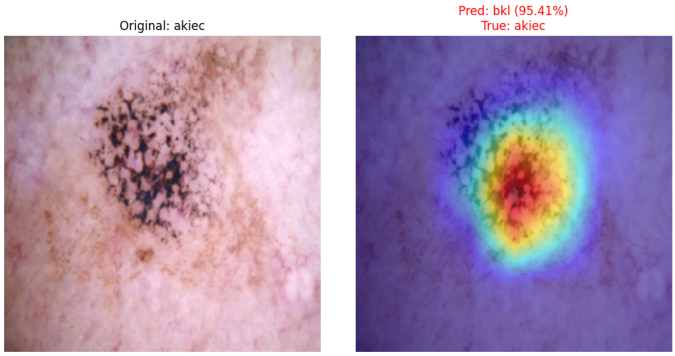


Fig. 7: Grad-CAM visualization of a failure case.

distinguish similar lesions. Besides, the focal loss dynamic weighting parameters might be experimentally implemented to reduce the remaining error rate of the minority classes. More advanced preprocessing techniques ought to also be developed to eliminate the pre training artifacts like hair and gel. Finally, one more aspect that we will carry out is the application of the model to a clinical workflow to identify its applicability in practice by dermatologists.

V. Conclusion

In this work, the long-standing problem of inequalities in classes during the medical image classification was finally overcome due to the use of the Class-Aware Multi-faceted Approach which was used to diagnose skin cancer on HAM10000. A methodical incorporation of focused information augmentation, the focal loss mechanism, and the Convolutional Block Attention Module allowed us to show that one can dramatically decrease the bias on the majority classes that is usually a limiting aspect of classic deep learning frameworks. The experimental findings were strong evidence of this increase because the proposed model recorded a Macro F1-Score of 0.7164 and a Balanced Accuracy of 0.6976 which significantly surpassed the EfficientNet-B0 baseline which had a tendency to record high false positive scores. Outside of the quantitative ones, the qualitative evaluation with Grad-CAM visualizations confirmed that the integration of attention mechanisms allows the model to concentrate on the most pertinent pathological patterns of the lesion, including pigment network abnormalities, instead of spurious artifacts, such as hair or gel bubbles. This correspondence to human diagnostic reasoning is an important aspect to fill a serious gap in the interpretability of AI systems, which makes them more acceptable in the context of clinical use. Although the research found that some components of the problem persist such as the misclassification of closely similar classes such as early-stage melanoma and the presence of nevi, the fact that the number of single error instances in our integrated strategy declined significantly compared to the vanilla architecture proves the strength of our approach. To summarize, the study confirms that a comprehensive set of data-level and architecture-level interventions is a solid solution to the development of reliable dermatological AI assistants that will provide a credible solution to the existing issue of matching the theoretical effectiveness of algorithms and the practical need to assess the patient with accuracy and explanation.

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